Team Spirit presents...

SunSpirit

sustainable technology for a cleaner tomorrow

The Problem:

Global Warming impacts **EVERYONE.**

Increased use of fossil fuels continues to deteriorate the ozone layer

Issues include:

- rising global temperatures and extreme weather
- damage to infrastructure and loss of agriculture
- decline in public health

Background

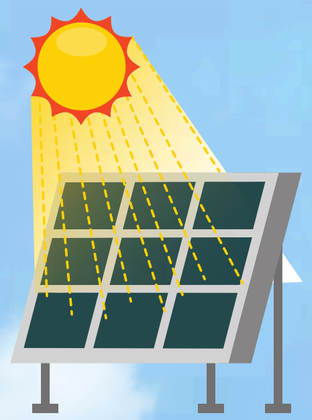
- Original use of the technology: improve the life quality in deep space for astronauts
- Through using a Sabatier method, waste carbon can be converted into fuel(In this case, methane).
- Modification -
- Our product follows the same direction as the original, with the exception of using the sabatier method. Instead, the device relies on water splitting.

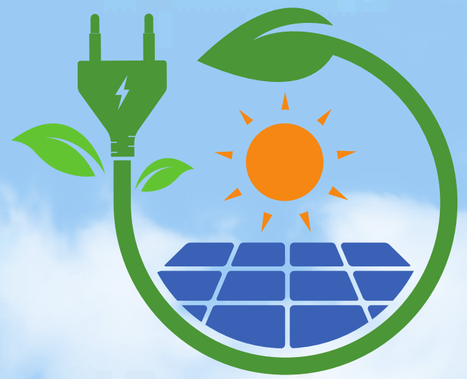
Our Solution

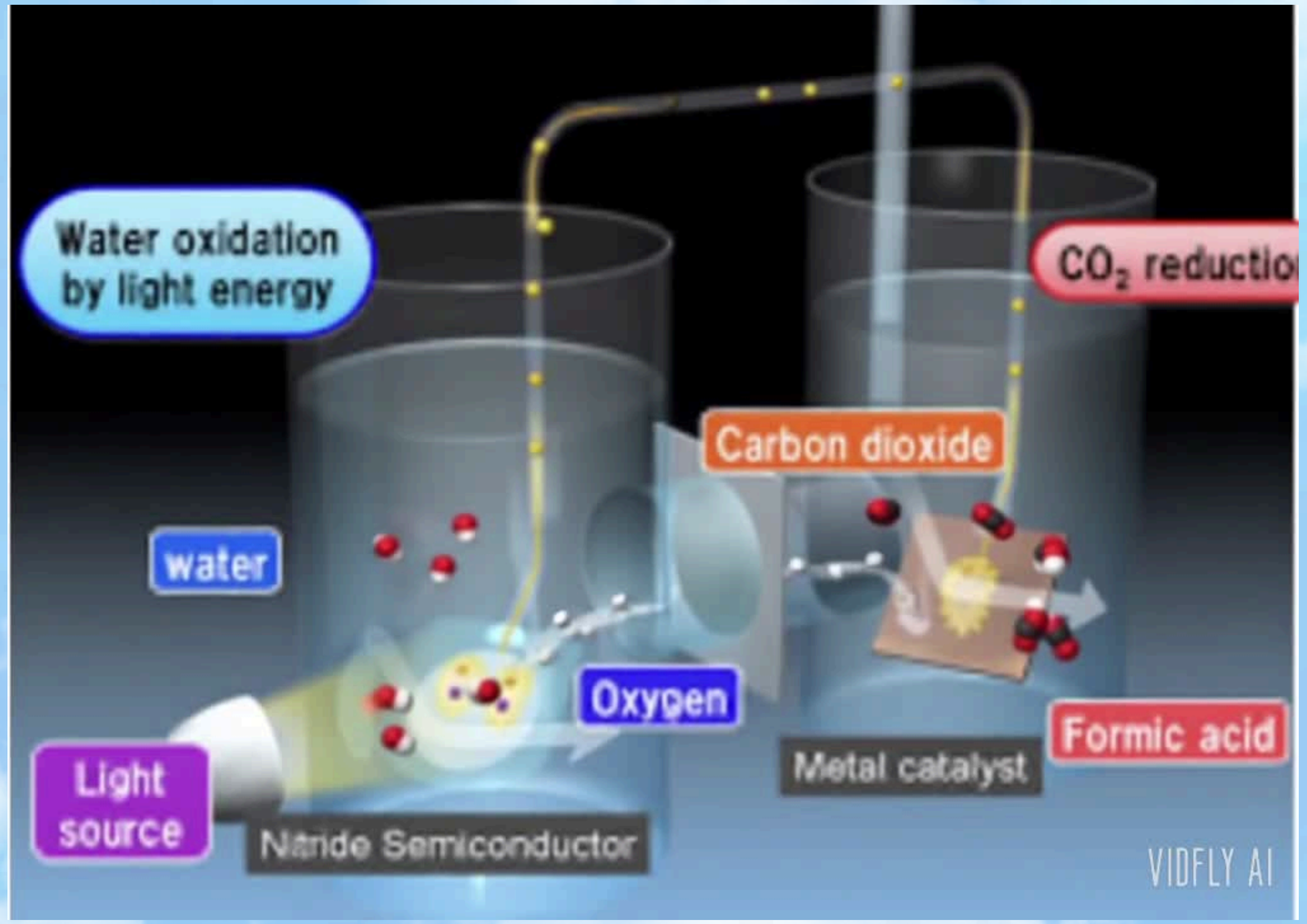
In order to efficiently use energy provided from natural resources, efforts can be made to incorporate solar energy into today's standard of living.

Our Solution

- The main technology our product plans to use involves the NASA patented tech for the conversion of CO_2,
- Non-Dispersion Infrared Sensors
- Artificial photosynthesis






Water oxidation
by light energy
CO_2 reductio
Carbon dioxide
water
Oxygen
Formic acid
Metal catalyst
Light
source
Nitride Semiconductor
VIDFLY AI

Costs

Component	Estimated Cost	Cost Strategy
Core Photoreactor	$5,000 - $15,000	Mass-produced vessels, automated TiO_2 coating
NDIR Sensors	$50 - $100	Integrated OEM chips (e.g., SCD40) at scale
Solar Power System	$3,000 - $7,000	Bulk Tier-1 panels (~$0.25/W), basic setup
Server & Monitoring	$20 - $50/month	Minimalist IoT data via low-cost cloud (e.g., AWS IoT Core)

based on our detailed breakdown, the estimated average cost to manufacture, integrade, and deploy a single unit is $12,495

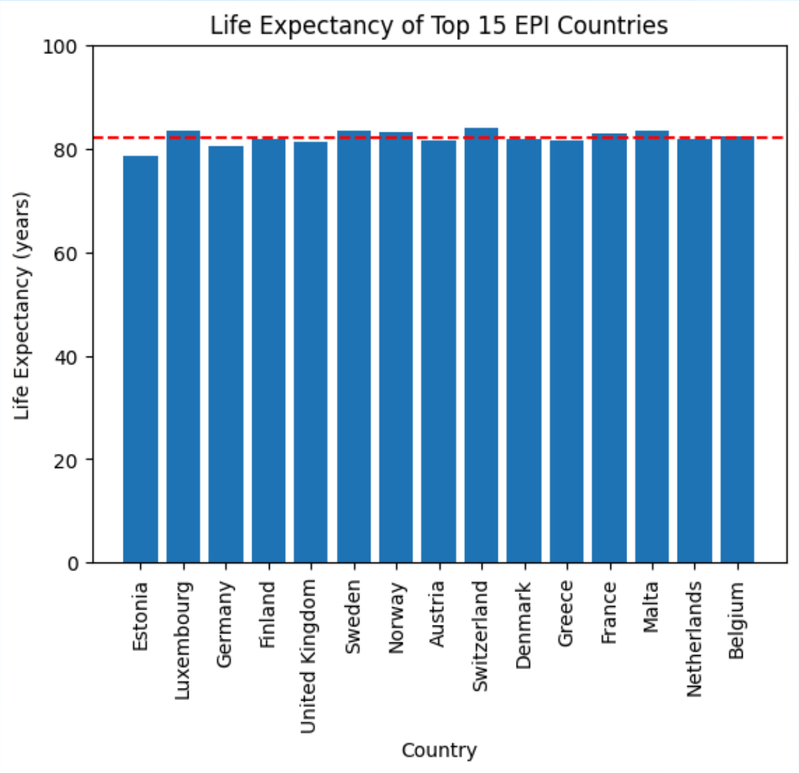
Government Solutions

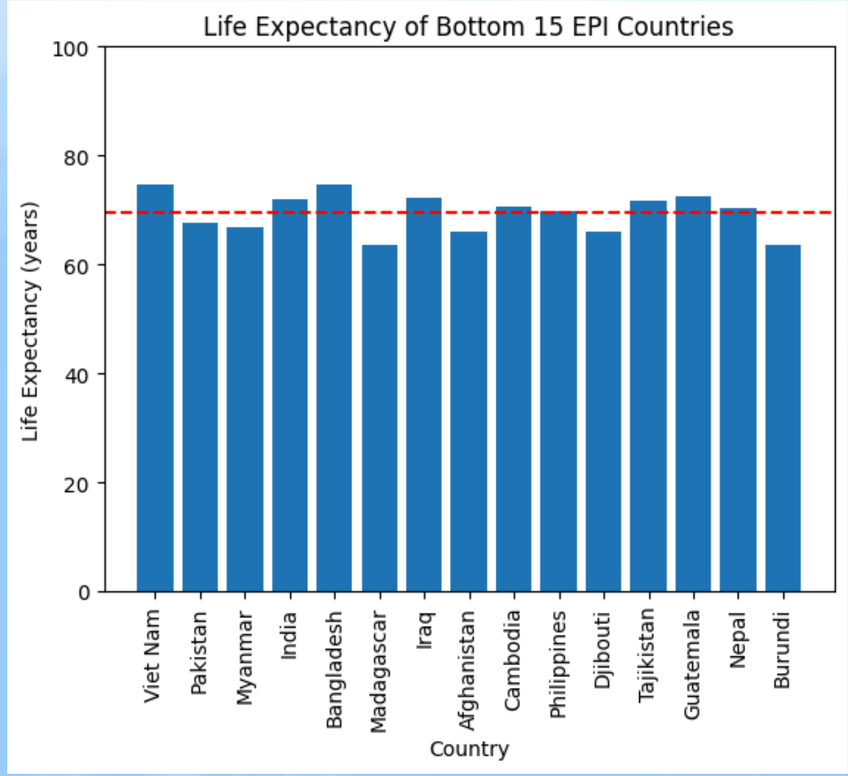
Households in the United States can save $34,000-$120,000 on their electric bills by going solar

India's solar city subsidies for installing solar powered infrastructure contributed to 5% of GDP growth in 2023

EU: clean energy initiatives accounted for nearly one-third of GDP growth in 2023

Countries that invest in sustainability have healthier, more productive workforces



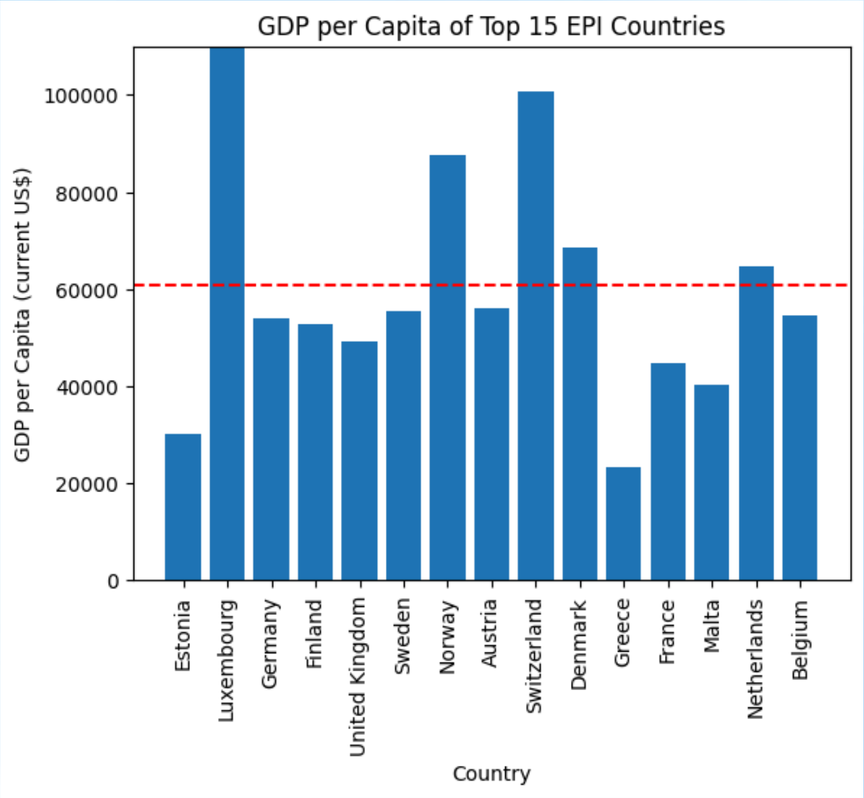


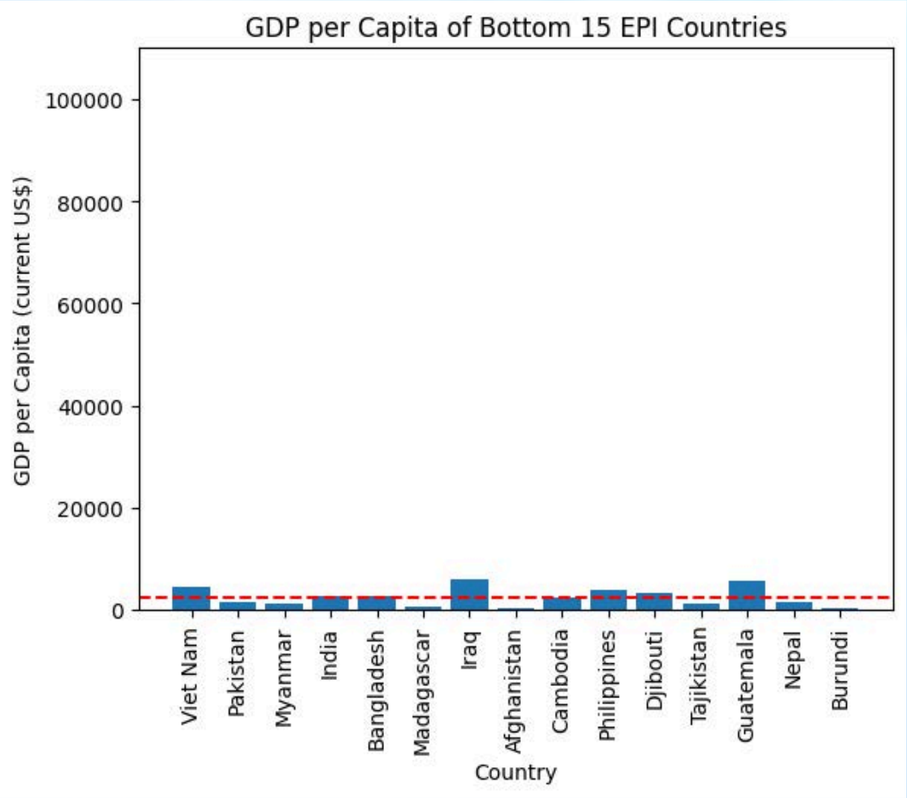
Average life expectancy: 82.1 years

Average life expectancy: 69.5 years

sources: Yale Environmental Performance Index, World Bank Index

Countries that invest in sustainability have healthier, more productive workforces



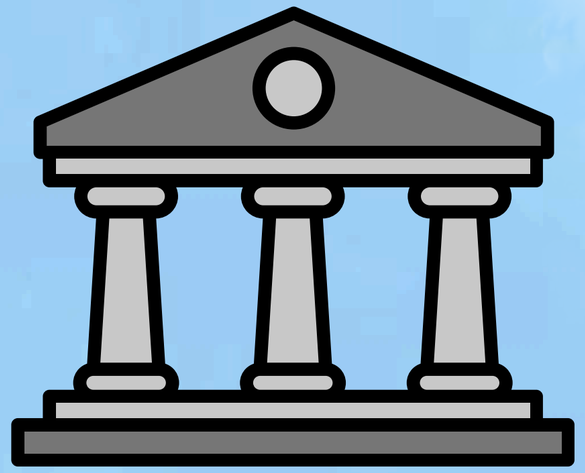


Average GDP per capita: $60881.57

Average GDP per capita: $2468.88

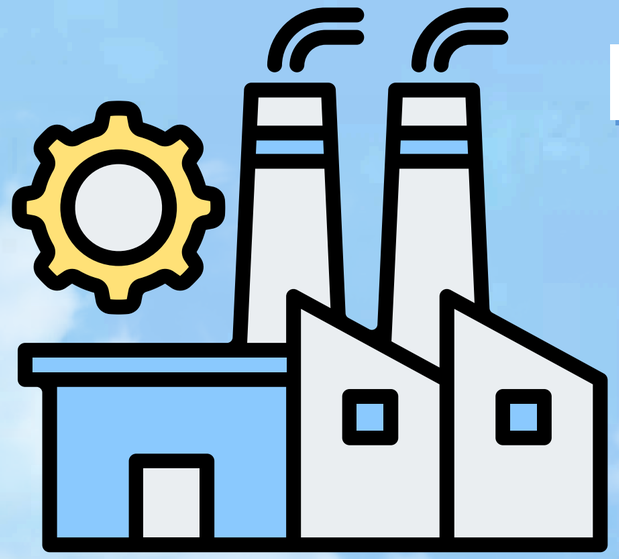
sources: Yale Environmental Performance Index, World Bank Index

Who are we selling to?



Governments:

- local
- federal



Industries:

- agriculture
- textile
- steel
- petroleum

Industrial Solutions

Tax Incentives

Improved Energy Efficiency

Improved Carbon Footprint

Laws and Incentives

NATIONAL LAWS -

Section 45Q Tax Credit for Carbon Sequestration: This is the most direct and valuable incentive for the SunSpirit project.

What it is: A tax credit awarded for every metric ton of qualified carbon oxide that is captured and utilized.

Value: For CO_2 that is utilized, the credit is valued at $60 per metric ton. This provides a direct, predictable revenue stream or cost offset for the industrial facility installing the technology.

STATE LAWS-

Georgia's Clean Energy Property Financing (PACE): The "Property Assesse Clean Energy" program allows property owners to finance the upfront cost of clean energy projects and then pay the costs back over time through a voluntary property tax assessment. This is an excellent financing tool which can be offered to potential clients to eliminate the barrier of high initial capital costs.

See it in action:
SunSpirit Pro!

https://sun-spirit.vercel.app

- Rising CO_2 needs solutions
- SunSpirit Pro: AI + sunlight → fuel
- HTML, CSS, JavaScript
- Solar collectors + CO_2 converters + sensors
- Emissions → energy, sunlight → sustainability